

THE WEST AFRICAN EXAMINATIONS COUNCIL

West African Senior School Certificate Examination
 for School Candidates

SC 2026

MATHEMATICS 1

1¼ hours

OBJECTIVE TEST

[40 marks]

Do not open this booklet until you are told to do so. While you are waiting, read and observe the following instructions. Write your name and index number in the spaces provided above.

1. Use 2B pencil throughout.
2. On the pre-printed answer sheet, check that the following details are correctly printed:
 - (a) In the space marked *Name*, check your surname followed by your other names.
 - (b) In the spaces marked *Examination*, *Year*, *Subject* and *Paper*, write 'WASSCE (SC)', '2026', 'MATHEMATICS', and '1' respectively.
 - (c) In the box marked *Index Number*, your index number has been printed vertically in the spaces on the left-hand side, and each numbered space has been shaded in line with each digit. Reshade each of the shaded spaces.
 - (d) In the box marked *Subject Code*, the digits 404112 are printed vertically in the spaces on the left-hand side. Reshade the corresponding numbered spaces as you did for your index number.
3. An example is given below. This is for a male candidate whose name is Tony Mensah ATIAH. His index number is 7102143958 and he is offering Mathematics 1.

THE WEST AFRICAN EXAMINATIONS COUNCIL
 ANSWER SHEET

PRINT IN BLOCK LETTERS	
Name: ATIAH TONY MENSAH	GHA
Examination: WASSCE (SC)	Year: 2026
Subject: MATHEMATICS	Paper: 1

INSTRUCTIONS TO CANDIDATES	
1. Use grade 2B pencil throughout.	
2. Answer each question by choosing one letter and shading like this: ☒ A ☐ B ☐ C ☐ D	
3. Erase completely any answer you wish to change.	
4. Leave extra spaces blank if the answer spaces provided are more than you need.	
5. Do not make any markings across the heavy black marks at the right hand edge of your answer sheet.	

INDEX NUMBER									
1	0	1	2	3	4	5	6	7	8
1	0	1	2	3	4	5	6	7	8
2	0	1	2	3	4	5	6	7	8
3	0	1	2	3	4	5	6	7	8
4	0	1	2	3	4	5	6	7	8
5	0	1	2	3	4	5	6	7	8
6	0	1	2	3	4	5	6	7	8
7	0	1	2	3	4	5	6	7	8
8	0	1	2	3	4	5	6	7	8

SUBJECT CODE									
1	0	1	2	3	4	5	6	7	8
1	0	1	2	3	4	5	6	7	8
2	0	1	2	3	4	5	6	7	8
3	0	1	2	3	4	5	6	7	8
4	0	1	2	3	4	5	6	7	8
5	0	1	2	3	4	5	6	7	8
6	0	1	2	3	4	5	6	7	8
7	0	1	2	3	4	5	6	7	8
8	0	1	2	3	4	5	6	7	8

For Supervisors only	
If candidate is absent shade this space	☐

Answer **all** the questions.

Mathematical tables may be used in **any** question. The use of non-programmable, silent and cordless calculator is allowed.

Each question is followed by **four** options lettered **A** to **D**. Find the **correct** option for **each** question and shade in **pencil**, on your answer sheet, the answer space which bears the same letter as the option you have chosen.

Give only **one** answer to **each** question. An example is given below.

The ages, in years, of four boys are 10, 12, 14 and 18. What is the average age of the boys?

- A. 12 years
- B. $13\frac{1}{2}$ years
- C. 13 years
- D. $12\frac{1}{2}$ years

The correct answer is $13\frac{1}{2}$ years, which is lettered **B**, and therefore answer space **B** would be shaded.

☐ A ☒ B ☐ C ☐ D

Think carefully before you shade the answer spaces; erase completely any answers you wish to change.

Do all rough work in this question paper.

Now answer the following questions.

1. Given that $\log_2 m = x$, $\log_2 n = x - 1$ and $\log_2 r = 2x$, find the value of $\log_2 (4mn^2r)$.
 - A. $4x^{-1}$
 - B. $4x$
 - C. $5x^{-1}$
 - D. $5x$
2. In a community of 100 assembly members, 30 are in **both** Works and the Finance Committees, and 45 are in the Finance Committee only. If **each** member is in **at least one** of the two committees, how many are in the Works Committee?
 - A. 15
 - B. 45
 - C. 55
 - D. 75

30
45
3. Given that the equation $27^{2x} = \left(\frac{1}{243}\right)^{2-3y}$ is written as $y = mx + c$, find the value of c .
 - A. $\frac{2}{3}$
 - B. $-\frac{2}{3}$
 - C. $\frac{3}{5}$
 - D. $-\frac{3}{5}$

27

4. Solve: $2\log_2 x = \log_2 4$.
- $x = 0$
 - $x = 1$
 - $x = 2$
 - $x = 4$
5. Okyere buys 7 pens and 10 exercise books for GH¢39.00. An exercise book costs GH¢0.50 more than a pen. Find the cost of an exercise book.
- GH¢ 1.50
 - GH¢ 2.00
 - GH¢ 2.50
 - GH¢ 3.00
6. Given that $\cos(2y) + \sin(20^\circ) = \tan(50^\circ)$ for $0^\circ \leq y \leq 90^\circ$, find, correct to one decimal place, the value of y .
- 15.9°
 - 22.4°
 - 37.7°
 - 45.3°
7. An equilateral triangle has side 8 cm. Find the area of the triangle.
- $4\sqrt{3} \text{ cm}^2$
 - $12\sqrt{3} \text{ cm}^2$
 - $16\sqrt{3} \text{ cm}^2$
 - $16\sqrt{5} \text{ cm}^2$
8. Afrah goes on a journey. She travels by taxi for 0.25 of the entire journey. She travels by motorbike for 0.6 of the entire journey and walks the remaining 1.83 km of the journey. Find the total distance of the entire journey.
- 12.2 km
 - 10.8 km
 - 7.4 km
 - 5.6 km
9. The complement of a given set is the set of all elements in the:
- set but not in the universal set.
 - set and also in the universal set.
 - universal set but not in the given set.
 - empty set but not in the given set.

10. Cynthia shared a 60 kg bag of rice among Ama, Bless and Issifu. Bless got **twice** as much as Ama and **thrice** as much as Issifu. If 5 kg of the rice was left after the sharing, how many kilograms did Bless get?
- 10
 - 20
 - 30
 - 40
11. Given that $x : y = 5 : 8$ and $y : z = 6 : 25$, find, in simplest form $x : y : z$.
- $5 : 8 : 125$
 - $5 : 48 : 100$
 - $15 : 24 : 100$
 - $30 : 48 : 25$
12. A 10-metre length wire increased in length to 10.245 m when its temperature increased. Calculate the percentage increase.
- 2.45%
 - 2.39%
 - 4.45%
 - 2.54%
13. A rectangle of dimension 6 cm by 4 cm is enlarged with a scale factor of 3. Find the area of the image of the rectangle.
- 216 cm^2
 - 72 cm^2
 - 36 cm^2
 - 24 cm^2
14. Bonsu and Kontor run at 3 km/h and 2 km/h respectively. If Bonsu run for 8 hours, how long will it take Kontor to run the same distance?
- 9 hours
 - 12 hours
 - 16 hours
 - 24 hours
15. Amuzu is 90 months younger than Konadu. If Konadu was born on 1st January 1950, when was Amuzu born?
- 1st January 1948
 - 1st July 1943
 - 1st July 1957
 - 1st June 1958
16. **Two thirds** of Ayensu's age is 5 years **more** than Barakisu's age. If Ayensu is 24 years old, find Barakisu's age.
- 29 years
 - 31 years
 - 19 years
 - 11 years

17. Find the values of y which satisfy the equation $y + \frac{5}{y-2} = \frac{9}{2}$.

- A. $y = -4, \frac{-5}{2}$
- B. $y = 4, \frac{-5}{2}$
- C. $y = 4, \frac{5}{2}$
- D. $y = -4, \frac{5}{2}$

18. The shape of the floor of a community centre is a regular polygon. If the sum of the interior angles of the floor is 1440° , find the number of sides.

- A. 6
- B. 8
- C. 10
- D. 12

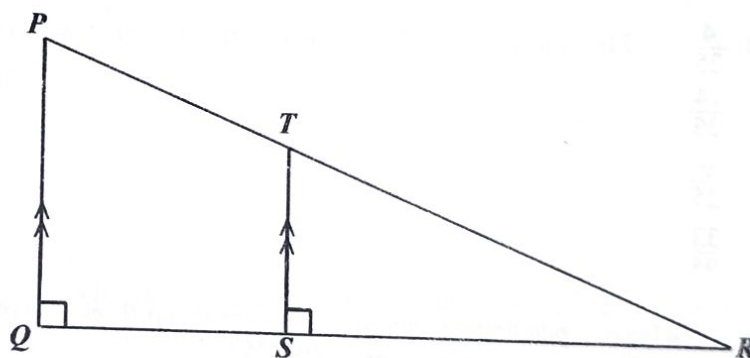
19. The starting monthly pay of a labourer in a certain school was GH¢ 1,800.00. At the beginning of **each** subsequent year GH¢ 240.00 is added to the previous year's pay. Find the labourer's pay at the beginning of the 11th year.

- A. GH¢ 2,400.00
- B. GH¢ 2,640.00
- C. GH¢ 4,200.00
- D. GH¢ 4,440.00

20. In a triangle, the largest interior angle is 4 times the size of the smallest angle. The other angle is 27° less than the largest angle. Find the largest angle.

- A. 69°
- B. 65°
- C. 92°
- D. 119°

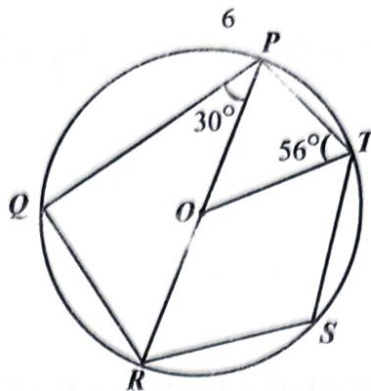
21.



NOT DRAWN TO SCALE

In the diagram, QSR and PTR are straight lines. $\overline{QP}$ is parallel to $\overline{ST}$, $|QR| = 20$ cm, $|QS| = 8$ cm and $|ST| = 9$ cm. Find $|QP|$.

- A. 5.4 cm
- B. 11 cm
- C. 15 cm
- D. 22.5 cm



NOT DRAWN TO SCALE

In the diagram, P, Q, R, S and T are points on a circle of centre O . $\angle QPR = 30^\circ$ and $\angle PTO = 56^\circ$.

Use this information to answer questions 22 and 23.

22. Find $\angle QRP$.

- A. 30°
- B. 60°
- C. 90°
- D. 120°

23. Find $\angle RST$.

- A. 94°
- B. 124°
- C. 150°
- D. 154°

24. A box contains 20 pens, all of the same size. 40% are black and the rest are blue. Two pens are selected at random, one after the other **without replacement**. Calculate the probability that **both** are blue.

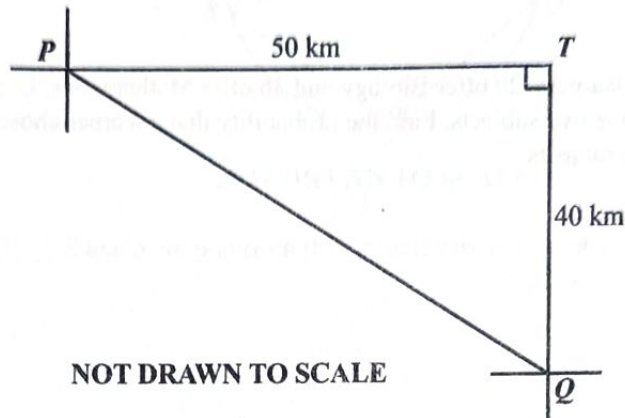
- A. $\frac{4}{25}$
- B. $\frac{14}{25}$
- C. $\frac{9}{25}$
- D. $\frac{33}{95}$

25. The costs (GH¢) of a pen and a pencil in a shop are P and R respectively. A student bought 15 pens and 5 pencils for Q . Express R in terms of P and Q .

- A. $R = \frac{Q + 15P}{5}$
- B. $R = \frac{Q - 15P}{5}$
- C. $R = \frac{Q - 5P}{15}$
- D. $R = \frac{Q + 15P}{15}$

26. A mason measured the height of a building to be 10.1^7 m. If the actual height of the building is 10 m, calculate the percentage error.
- A. 10%
 B. 1.01%
 C. 1.0%
 D. 0.1%

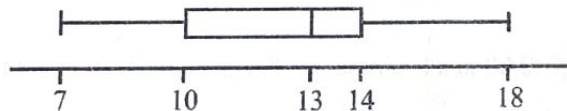
27.



The diagram shows two communities P and Q in a plane. Find the bearing of Q from P .

- A. 039°
 B. 051°
 C. 129°
 D. 309°

28.



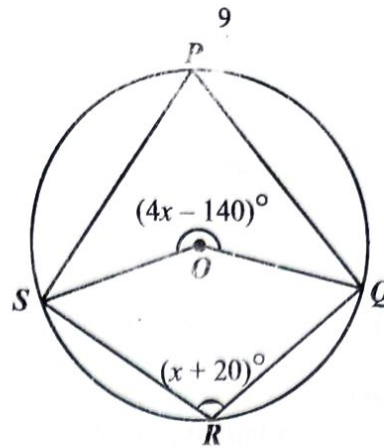
NOT DRAWN TO SCALE

The box plot shows the distribution of marks scored by 60 students in a test. Find the interquartile range.

- A. 10
 B. 14
 C. 4
 D. 3
29. A farmer hired labourers to weed a circular farm of radius 210 m. If each labourer weeded $9,240 \text{ m}^2$, how many labourers were hired? [Take $\pi = \frac{22}{7}$]
- A. 7
 B. 14
 C. 15
 D. 44

30. Priscilla climbed a coconut tree which was slanting at 35° to the horizontal ground. She plucked fruit and dropped it vertically to the ground through a distance of 20.5 m. How far from the ground, to the nearest metre, did Priscilla climb along the tree?
- 12 m
 - 17 m
 - 25 m
 - 36 m
31. In a class of 50 learners, 20 offer Biology and 36 offer Mathematics. Each of the learners offer **at least one** of the two subjects. Find the probability that a learner chosen at random from the class offer **both** subjects.
- $\frac{22}{55}$
 - $\frac{5}{6}$
 - $\frac{7}{10}$
 - $\frac{3}{25}$
32. The diameter of a wheel of a bicycle is 84 cm. A cyclist rides the bicycle over a distance of 46.2 km. Calculate the number of revolutions made. [Take $\pi = \frac{22}{7}$]
- 35,000
 - 17,500
 - 8,750
 - 4,375
33. A farmer invested GH¢ 6,000.00 in a business at 5.5% per annum simple interest. Find the amount at the end of 8 years.
- GH¢ 11,280.00
 - GH¢ 10,800.00
 - GH¢ 8,640.00
 - GH¢ 8,400.00
34. The Assembly member of a community observed that boys in the community **do not** go to school on market days. Which type of data collection tool was used?
- Primary and qualitative
 - Primary and quantitative
 - Secondary and qualitative
 - Secondary and quantitative
35. At the beginning of 2022, Mr Appiadu's house was worth GH¢ 240,000.00. The value of his house increased by 4% **each** year. Find the value at the beginning of 2025.
- GH¢ 249,600.00
 - GH¢ 259,584.00
 - GH¢ 269,967.36
 - GH¢ 280,766.05

36.



NOT DRAWN TO SCALE

In the diagram, P , Q , R and S are points on the circumference of a circle, centre O . Find the value of x .

- A. 70°
- B. 80°
- C. 90°
- D. 95°

The data shows the marks scored by eight students in a test. The ordered scores are: $(x + 3)$, 9, $(2x + 3)$, 13, 15, 16, 18, $(x^2 - 5)$ and the range is 12.

Use the information to answer questions 37 to 39.

37. Find the value of x .

- A. 3
- B. 4
- C. 5
- D. 6

38. Calculate the mean score of the test.

- A. 13
- B. 14
- C. 15
- D. 16

39. If a student is selected at random, find the probability that he/she scores **at least** 15.

- A. $\frac{1}{2}$
- B. $\frac{3}{4}$
- C. $\frac{1}{4}$
- D. $\frac{7}{8}$

Turn Over

SC 4042
WASSCE 2026
MATHEMATICS 2
Essay
2½ hours

2

Name

Index Number

THE WEST AFRICAN EXAMINATIONS COUNCIL

West African Senior School Certificate Examination
for School Candidates

SC 2026

MATHEMATICS 2
ESSAY

2½ hours

Write your name and index number in ink in the spaces provided above.

*Answer **ten** questions in all. All the questions in Section A and **five** questions from Section B.*

*In **each** question, all necessary details of working, including rough work, **must** be shown with the answer.*

Give answers as accurately as data and tables allow.

Graph papers are provided for your use in the examination.

The use of non-programmable, silent and cordless calculator is allowed.

2
SECTION A
[40 marks]

Answer all the questions in this section. All questions carry equal marks.

1.
 - (a) Ato bought 150 kg of fish for GH¢ 2,250.00. At what price should a kilogram of the fish be sold to make **at least** 20% profit?
 - (b) The population growth of a community follows the model $P = 20,000(2.72)^{0.03t}$, where t is the number of years after 2020. In how many years will the population reach 30,000?

2. A Pharmacist asked a patient to take 5 ml every 8 hours of a 100 ml mixture. The patient began to take the mixture on Tuesday at 10:00 am.
 - (a) On what day will the mixture get finished?
 - (b) At what time will the patient take the last mixture?

3. The perimeter of the base of a cylindrical water tank is 220 cm and the height is 56.5 cm. Calculate, correct to **two** decimal places:
 - (a) the total surface area;
 - (b) in litres, the quantity of water the tank can hold. [Take $\pi = 3.142$]

4. Two fair six-faced dice are rolled at the same time.
 - (a)
 - (i) State whether the two events are independent or not.
 - (ii) Give reason for the answer in (a)(i).
 - (b) Find the probability that:
 - (i) the **first** die showed a 2, and the **second** die showed an odd number;
 - (ii) **both** dice showed even numbers.

5. The size of a farmland that can be worked on depends on a fixed number of permanent labourers and a variable number of hired labourers. For a 14 acre farm, 10 labourers are hired and a 59 acre farm, 100 labourers are hired. Find the:
 - (a) number of permanent labourers;
 - (b) size of a farm land which would require 80 hired labourers.

3
SECTION B
[60 marks]

This section consists of parts I, II, III and IV.

Answer five questions only from this section with at least one question from each part.

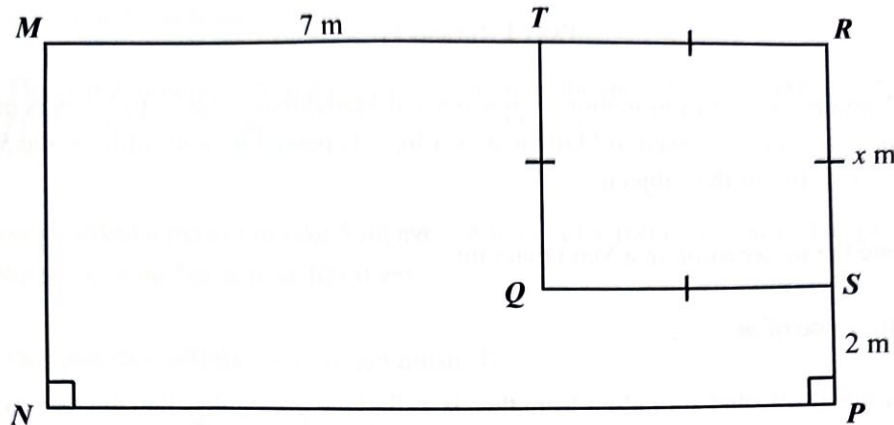
PART I (Strand 1)

6. In a class, 50 learners took an examination in Science and Mathematics. $(m^2 - 6)$ learners passed in Science, $(6m + 4)$ learners passed in Mathematics, $(4m + 1)$ passed in both subjects and 5 of them did **not** pass in any of the subjects.
- (a) Illustrate the information in a Venn diagram.
- (b) Find the value of m .
- (c) If a learner is selected at random from the class, find the probability that the learner passed in:
- (i) **only** one subject;
- (ii) Science but **not** Mathematics.
7. (a) In a fun club, there were 100 **more** males than females. When 20 **more** females joined the club, the ratio of males to females became 5 : 3. Find the number of females **initially** in the club.
- (b) Rosemond can trim a hedge in 12 days and Lucy can do it in 15 days. If they are joined by Alice, they can complete the work in 4 days. How long will it take Alice alone to complete the same work?

PART II (Strand 2)

8. (a) The sum of three natural numbers is 70. The **first** number is **two-thirds** the **second** number. The **third** number is the sum of the **first** and **second** numbers. Find the numbers.
- (b) A telecommunication company offers 6 hours of talk time and 5GB of data for GH¢ 59.00. It also offers 24 hours of talk time and 16GB of data for GH¢ 208.00.
- (i) Write a mathematical model for the information.
- (ii) Find the cost of:
- (I) one hour of talk time;
- (II) one GB of data.
9. (a) Given that $a = u + \frac{V}{b}$ and $R = u + \frac{V}{a}$, express b in terms of u , V and R .
- (b) A woman shared GH¢ 7,280.00 among all her children. Starting from the youngest she gave GH¢ 20.00, GH¢ 60.00, GH¢ 180.00 and so on. How many children does the woman have?

10. (a) A point, Q is translated by the vector $\begin{pmatrix} 15 \\ -16 \end{pmatrix}$ onto the image $(10, -4)$. Find $Q(x, y)$.
- (b)



NOT DRAWN TO SCALE

In the diagram, $MNPR$ is a rectangular lawn and $TQSR$ is a square of side x m to be used as flower bed. $|MT| = 7$ m, $|SP| = 2$ m and the total area of the lawn is 50 m^2 .

Find the:

- (i) value of x ;
 - (ii) length and width of the lawn.
11. On one particular morning a man and his daughter left their house. The man went due South of the house and his daughter went due East of the house. The average speed, in km/h, of the man was **one and half** times the daughter's average speed. After 2 hours, the distance between them was $\frac{22}{5}\sqrt{13}$ km.
- (a) Illustrate the information in a diagram.
 - (b) Calculate the distance of the man from the house.
 - (c) Find the bearing of the man from his daughter.

PART IV (Strand 4)

12. (a) The marks scored by some learners in a test are: 9, 8, 6, 7, 4, 5 and 4. Calculate:
- (i) correct to **two** decimal places, the mean mark;
 - (ii) the median.
- (b) A bag contains green and red marbles, all of the same size. The number of red is 12 **more than** the number of green. The probability of selecting at random a green marbles is $\frac{1}{5}$. Find the total number of marbles in the bag.

13. (a) The scores of some students in a test are: 16, 11, 8, 15, 12, 14, 12, 7, 16, 10 and 10. Find the semi-interquartile range.
- (b) The data shows the ordered marks obtained by some candidates in an interview:
30, 43, 47, $(y^2 + 14)$, $(10y - 3)$, 72, 84 and 88.
- (i) If the median mark is 65, find the value of y .
- (ii) Given that 37.5% of the candidates passed the interview, determine the pass mark.

END OF PAPER